Gas Generator Tachometer Generator – Maintenance Practices

1. Adjustment/Test

A. Bench Test Tachometer Generator

Refer to for details.

Equipment required:

- Variable speed motor
- Drive speed indicator
- Tachometer generator output indicator (percent rpm)
- Mounting fixture and wiring
- Motor control switch
- HI–RPM–LOAD (Three 40 ohm, Y connected resistances)
- LO–RPM–LOAD (Three 20 ohm, Y connected resistances)
- Load ON–OFF switch (Two required)
- Multimeter

1) Continuity Test:

a) Set multimeter to OHMS R x 1.

b) Measure the continuity resistance between each pair of stator windings A–B, A–C and B–C (C = Case). The resistance between each pair of windings must be less than 50 ohms and must be within 2.0 ohms of each other.

2) Rotation Test:

a) Connect tachometer generator to output indicator as shown in Figure 1.

b) Operate motor and check output indicator pointer moves in a clockwise direction as motor speed is increased.

Note Tachometer generator is rotated in a counterclockwise direction when viewed from shaft end.

c) Shutdown motor.

3) Voltage Output Test:

a) Disconnect wiring from output indicator (refer to Para 1.A.2).a)) and connect to test point terminals as shown in Figure 1.

b) Ensure LO–RPM–LOAD switch is OFF, and select HI–RPM–LOAD switch to ON.

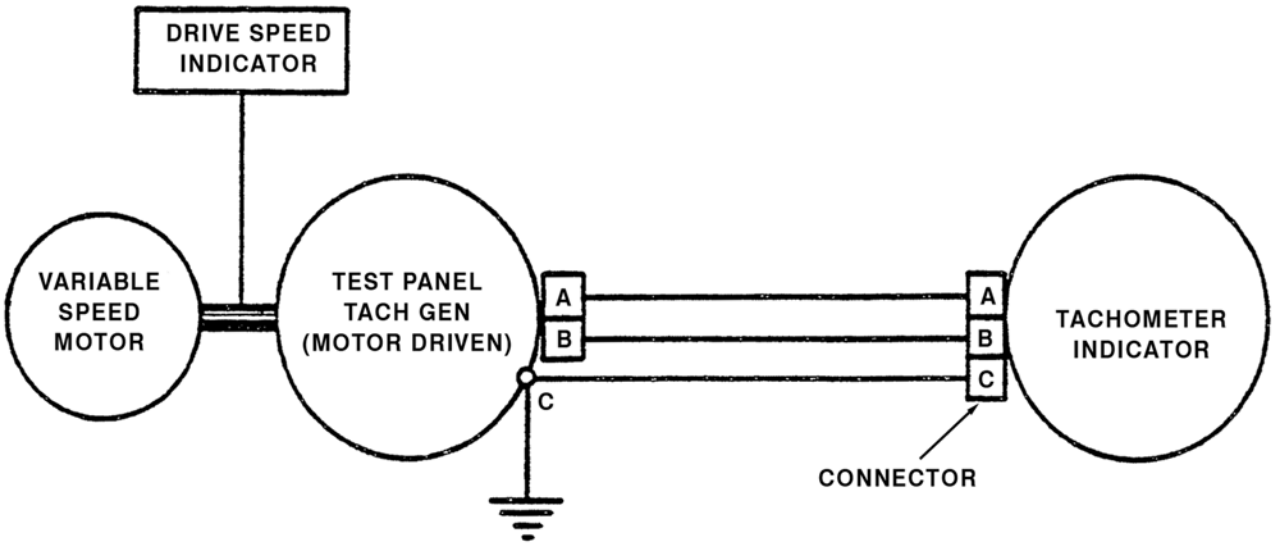
c) Operate motor and increase speed to 4200 rpm (drive speed indicator).

d) Set multimeter to 30 volts AC range and measure voltage outputs at the test point terminals across each of the phases A–B, A–C and B–C. The voltage across each phase must be 21 ± 0.5 volts AC.

e) Decrease motor speed to 1000 rpm, select HI–RPM–LOAD switch to OFF and LO–RPM–LOAD switch to ON.

f) Repeat Para 1.A.3).d) and check that phase voltages are not less than 3.5 volts AC.

g) Remove tachometer generator from test bench.



NOTE: TACHOMETER GENERATOR (TACH GEN) IS ROTATED IN A CLOCKWISE DIRECTION WHEN VIEWED FROM THE SHAFT END

Test Rig Layout Schematic
Figure 1